

Requirement Analysis Document

Online Shopping Platform (E-Commerce System)

1. Introduction

For this study, we looked at how online shopping platforms like Amazon and Flipkart work, and tried to list out the main activities that happen when a customer uses such a website or app. Based on this, we have prepared the following requirement analysis for a proposed online shopping system.

2. Purpose of the System

The main purpose of this system is to let customers shop online without visiting a physical store. It should allow them to search for products, place orders, pay for them, and keep track of delivery, while also giving the seller/admin a way to manage the products and orders.

3. Scope

The system covers the customer side (registration, browsing, cart, checkout, payment, tracking) as well as basic admin/seller functions like adding products and updating stock. Things like advanced logistics or supplier management are not covered here, since the focus is on the core shopping experience.

4. Who Uses the System

- Customer** - browses products, places orders, pays, and tracks delivery.
- Seller / Admin** - adds and manages products, views orders, updates stock.
- Delivery Person** - updates the shipment status as the order is delivered.
- Payment Gateway** - handles the online payment part securely.

5. Major Activities of the System

These are the main activities we identified, along with what the system should be able to do for each one.

5.1 Customer Registration and Login

- A new user should be able to sign up using name, email, phone number and a password.
- Existing users should be able to log in with their email/phone and password.
- There should be an option to reset the password if it is forgotten.
- Users should be able to view and edit their profile and saved addresses.

5.2 Product Browsing and Search

- Users should be able to look at products by category, like electronics, clothes, groceries, etc.
- There should be a search bar to find products by name or keyword.
- Users should be able to filter/sort by price, rating or popularity.
- Each product should have its own page with price, description, images and reviews.

5.3 Cart and Order Placement

- Users can add products to a cart and remove them if they change their mind.
- Users can change the quantity of items in the cart before checkout.
- During checkout, the user selects a delivery address and confirms the order.
- Once placed, the order should get a unique order ID so it can be tracked later.

5.4 Payment Processing

- The user should be able to choose how to pay - card, UPI, net banking or cash on delivery.
- Online payments should go through a secure payment gateway.
- After payment, the user should get a confirmation message and a receipt/invoice.
- If a payment fails, the system should let the user know and allow them to try again.

5.5 Shipment and Order Tracking

- The order status should update as it moves along - placed, packed, shipped, out for delivery, delivered.
- Customers should be able to track their order using the order ID.
- The system should send updates through email or SMS at each stage.
- Customers should be able to request a cancellation or return within a certain number of days.

5.6 Admin / Seller Side

- Sellers should be able to add new products or edit/remove existing ones.
- Sellers need a way to keep the stock/inventory numbers updated.
- Admins should be able to see incoming orders and their status.
- It would help to have a simple report of sales/orders for reference.

6. Other Requirements to Keep in Mind

Apart from the main features above, a few general things matter for a system like this:

- The website/app should be easy to use, even for people who are not very tech-savvy.
- Pages should load quickly, especially search results and the checkout page.
- Customer details and payment information need to be kept secure at all times.
- The platform should ideally be available all day, every day, without much downtime.
- It should be able to handle more users and orders as the platform grows.

7. Conclusion

Overall, the idea is to give customers an easy way to browse, order, pay and track their products, while also giving the seller side a simple way to manage things on their end. This document should be a good starting point for designing and building the actual system.